Advanced Algorithms
COMP4121

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Recap on Fourier series and integrals

Finitely dimensional vector spaces

- Vectors: $x = \langle x_0, x_1, \ldots, x_{n-1} \rangle$
- Scalar product:

$$\sum_{k=0}^{n-1} x_k \overline{y_k}$$

- Basis:

$$\{\varphi_0, \ldots, \varphi_{n-1}\}; \qquad \varphi_k = \frac{1}{\sqrt{n}} \left(e^{i\,\frac{2\pi}{n}\cdot k\cdot 0}, e^{i\,\frac{2\pi}{n}\cdot k\cdot 1}, \ldots, e^{i\,\frac{2\pi}{n}\cdot k\cdot m}, \ldots, e^{i\,\frac{2\pi}{n}\cdot k\cdot (n-1)}\right)$$

- (finite) Fourier series (DFT):

$$\widehat{x}_k = \langle x, \varphi_k \rangle = \frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} x_i e^{-\,i\,\frac{2\pi}{n}\cdot k\cdot i}$$

$$x = \sum_{k=0}^{n} \langle x, \varphi_k \rangle \varphi_k = \sum_{k=0}^{n} \widehat{x}_k \varphi_k$$

$$x_m = \frac{1}{\sqrt{n}} \sum_{k=0}^{n} \widehat{x}_k e^{i\,\frac{2\pi}{n}\cdot k\cdot m}$$

2π-periodic functions

- Vectors: 2π-periodic (piece-wise continuous) functions $f(x)$
- Scalar product:

$$\langle f, g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\overline{g(x)}$$

- Basis:

$$\{\dots, \varphi_{-k}(x), \dots, \varphi_0(x), \dots, \varphi_k(x), \dots\}; \qquad \varphi_k(x) = e^{i\,k\,x};$$

- Fourier series:

$$\widehat{f}_k = \langle f, \varphi_k\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-\,i\,k\,x}dx$$

$$f(x) = \sum_{k=-\infty}^{\infty} \langle f, \varphi_k\rangle \varphi_k(x) = \sum_{k=-\infty}^{\infty} \widehat{f}_k\, e^{i\,k\,x}$$

- Note: every function $f(x)$ which is zero outside $(-\pi, \pi)$ can be extended to a 2π periodic function by concatenating shifted copies of $f(x)$.

L^2 functions

- Vectors: functions of "finite energy", i.e., satisfying

$$\int_{-\infty}^{\infty} |f(t)|^2 dt < \infty$$

- Scalar product:

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(t)\overline{g(t)} dt$$

- Heuristics for Fourier integral: "infinitely refined basis": $\{e^{i\,\omega\,t} \;: \omega \in \mathbb{R}\}$
- "Infinitely refined Fourier coefficients":

$$\widehat{f}(\omega) = \langle f(t), e^{i\,\omega\,t} \rangle = \int_{-\infty}^{\infty} f(t) e^{-\,i\,\omega\,t} dt$$

- "Infinitely refined Fourier series":

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \langle f(t), e^{i\,\omega\,t} \rangle e^{i\,\omega\,t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{f}(\omega) e^{i\,\omega\,t} d\omega$$

π band limited signals of finite energy

- Vectors: L^2 functions with Fourier transform $\widehat{f}(\omega) = 0$ for all $\omega \notin (-\pi, \pi)$
- Thus, for such signals

$$f(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \widehat{f}(\omega) e^{i\,\omega\,t} d\omega$$

- Since $\widehat{f}(\omega) = 0$ for all $\omega \notin (-\pi, \pi)$, we can expand $\widehat{f}(\omega)$ into Fourier series in the usual basis for 2π periodic functions $\mathcal{B} = \{e^{i\,k\,\omega} \ : \ k \in \mathbb{Z}\}$
- Thus,

$$\widehat{f}(\omega) = \sum_{k=-\infty}^{\infty} \langle \widehat{f}, e^{i\,k\,\omega} \rangle \, e^{i\,k\,\omega}$$

- But

$$\langle \widehat{f}, e^{i\,k\,\omega} \rangle \ = \frac{1}{2\pi} \int_{-\pi}^{\pi} \widehat{f}(\omega) e^{-\,i\,k\,\omega} d\omega = f(-k)$$

- Thus,

$$\widehat{f}(\omega) = \sum_{k=-\infty}^{\infty} f(-k) \, e^{i\,k\,\omega} = \sum_{k=-\infty}^{\infty} f(k) \, e^{-\,i\,k\,\omega}$$

Finitely dimensional vector spaces

- We can now substitute

$$\widehat{f}(\omega) = \sum_{k=-\infty}^{\infty} f(k)\, e^{-i\,k\,\omega} \qquad \text{into} \qquad f(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \widehat{f}(\omega) e^{i\,\omega\,t} d\omega$$

and obtain

$$\begin{aligned}
f(t) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \widehat{f}(\omega) e^{i\,\omega\,t} d\omega \\
&= \frac{1}{2\pi}\int_{-\pi}^{\pi} \left(\sum_{k=-\infty}^{\infty} f(k)\, e^{-i\,k\,\omega} \right) e^{i\,\omega\,t} d\omega \\
&= \sum_{k=-\infty}^{\infty} f(k)\, \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-i\,k\,\omega} e^{i\,\omega\,t} d\omega \\
&= \sum_{k=-\infty}^{\infty} f(k)\, \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i\,(t-k)\,\omega} d\omega \\
&= \sum_{k=-\infty}^{\infty} f(k)\, \frac{\sin \pi(t-k)}{\pi(t-k)} \\
&= \sum_{k=-\infty}^{\infty} f(k) \operatorname{sinc} \pi(t-n)
\end{aligned}$$

Representation of band limited signals of finite energy

- Thus, we get the following isometric spaces, with their bases and the corresponding Fourier series:

frequency domain:

$$\widehat{f}(\omega) \qquad \{e^{i\,n\omega} \;:\; n \in \mathbb{Z}\} \qquad \widehat{f}(\omega) = \sum_{n=-\infty}^{\infty} f(n)\, e^{-\,i\,n\omega}$$

time domain:

$$f(t) \qquad \{\operatorname{sinc} \pi(t-n) \;:\; n \in \mathbb{Z}\} \qquad f(t) = \sum_{n=-\infty}^{\infty} f(n) \operatorname{sinc} \pi(t-n)$$

sequences from l^2

$$\vec{f} = \{f(n)\}_{n\in\mathbb{Z}} \qquad \{e_n = (\ldots 0, 1, 0 \ldots) \;:\; n \in \mathbb{Z}\} \qquad \vec{f} = \sum_{n=-\infty}^{\infty} f(n)\, e_n$$

- Isometries:

$$\|f(t)\|^2 = \int_{-\infty}^{\infty} |f(t)|^2 dt \;=\; \|\widehat{f}(\omega)\|^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} |f(\omega)|^2 d\omega \;=\; \|\vec{f}\|^2_{l_2} = \sum_{n=-\infty}^{\infty} |f(n)|^2$$

Action of linear operators on band limited signals

- Let $L[\operatorname{sinc} \pi\, t] = l(t)$, and let $L[\widehat{\operatorname{sinc} \pi\, t}] = \widehat{l}(\omega)$; then

<u>time domain:</u>

$$f(t) = \sum_{n=-\infty}^{\infty} f(n) \operatorname{sinc} \pi (t-n) \qquad \Rightarrow \qquad L[f](t) = \sum_{n=-\infty}^{\infty} f(n)\, L[\operatorname{sinc} \pi\, u](t-n)$$

$$f(t) \qquad \Rightarrow \qquad \widehat{L[f]}(t) = \int_{-\infty}^{\infty} f(u) l(t-u) du = \int_{-\infty}^{\infty} f(t-u) l(u) du$$

<u>frequency domain:</u>

$$\widehat{f}(\omega) \qquad \Rightarrow \qquad \widehat{L[f]}(\omega) = \widehat{f}(\omega)\, \widehat{l}(\omega)$$

<u>domain of discrete samples (sequences from l^2):</u>

$$\vec{f} = (f(n) \ :\ n \in \mathbb{Z}) \qquad \Rightarrow \qquad L[\vec{f}] = (L[f](n) \ :\ n \in \mathbb{Z})$$

where

$$L[\vec{f}](n) = \sum_{k=-\infty}^{\infty} f(n-k) l(k)$$

Action of linear operators on band limited signals of finite energy

- In practice both $f(n)$ and $l(k)$ have only finitely many samples.
- Thus,

$$L[\vec{f}](n) = \sum_{k=-\infty}^{\infty} f(n-k)l(k)$$

becomes

$$L[\vec{f}](n) = \sum_{k=-M/2}^{M/2} f(n-k)l(k) \quad \text{for all} \quad n = 0 \dots K$$

- M is usually less than 1000 but K can be tens of millions of samples for a single song, so we split $\vec{f}$ into frames of size equal to the size of $\vec{l}$ and compute the convolution using the FFT and then put the resulting pieces of the output signal together by aligning them appropriately.